Lecture 4

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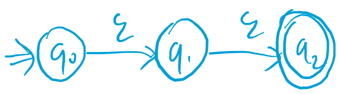
Problem Set · 2

1. Constructing a DFA with 1 element.



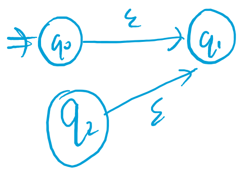
NFA

DFA. ($<k$).
$\Rightarrow 3$

2. Minimal NFA's are not unique.
↓
Non-deterministic

w·NFA. same no. of states recognize same lang.

3. $Q = \{q_0, q_1, q_2\}$



$A = \{q_0\}$
$B = \{q_2\}$.

$A \cap B = \emptyset$. $E(A \cap B) = E(\emptyset) = \emptyset$.

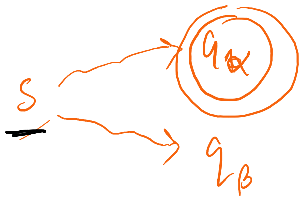
$E(A) \cap E(B) = \{q_1\}$

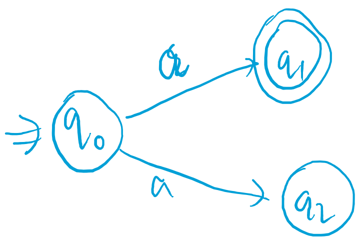
4. In DFA, a str. is either accepted or rejected.
$\Rightarrow q_0 \to \dots \to q_n$ where either $q_n \in F$ or $q_n \notin F$.

F → Q/F.
rejected accepted.

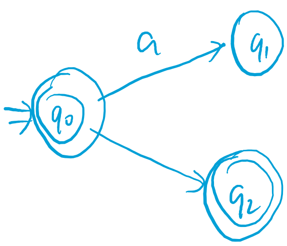
Complement DFA.

str in NFA $\xrightarrow{\varepsilon t}$ $\xrightarrow{\varepsilon t}$





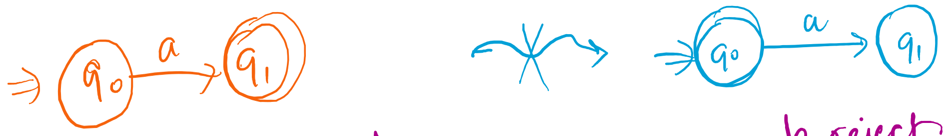
a accepted



a accepted

a accepted

$\{a,b\} \subseteq \Sigma$



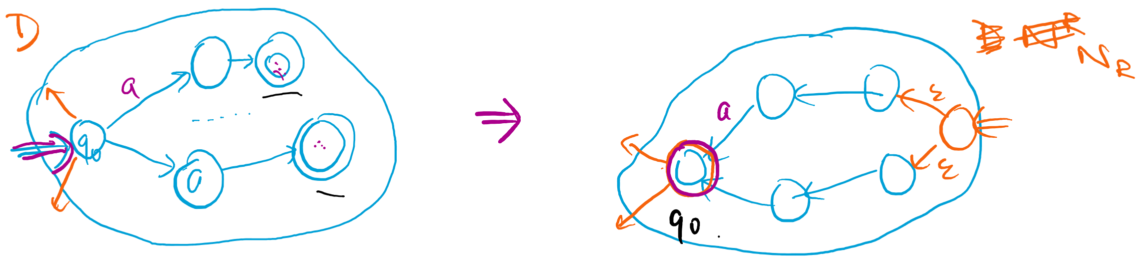
b reject

b reject

5. Thm. L is regular $\Rightarrow$ L^R is regular. (Closure).

$\exists$ a DFA $= (Q, \Sigma, \delta, q_0, F)$ recognize L.

$$\left[L^R = \{ a_n a_{n-1} \cdots a_1 = w^R \mid w = a_1 a_2 \cdots a_n \in L \},\ a_i \in \Sigma\ \forall i. \right]$$



$$\delta'(q_i, a) = \begin{cases} q_j; & \text{if } \exists q_j \text{ s.t. } \delta(q_j, a) = q_i \\ \phi; & \text{otherwise.} \end{cases}$$

$q_i \neq q_0$

$+\ \delta'(s, \varepsilon) = F$

$N = (Q, \Sigma, \delta', s, \{q_0\})$.

N recognizes L^R.

$\Rightarrow L^R$ is regular.

Pumping Lemma : For every regular lang. A.

P_1: $\exists p$ (pumping length) s.t. if $(s \in A)$ and $(|s| \geq p.)$

P_2: then $s = xyz$ where

1. $xy^iz \in A \quad \forall i \geq 0$.

Finite regular.

Regular

1. $xy^iz \in A \quad \forall i \geq 0.$

2. $y \neq \varepsilon.$

3. $|xy| \leq p.$

Regular

⇒ Pumping lemma.

(Necessary).

$(\exists p)(\forall s \in L) \longrightarrow (\forall s \in L)(\exists p)$

$s \to L \qquad p(s)$

$P_1 \subseteq P_2 \Rightarrow P_2$ is weaker

$s \in A$ ↳ finite language regular.

$s = a_1 a_2 \ldots a_n.$

$p_{(s)} \leq n \qquad x, y$

$s = \underbrace{a_1 \ldots}_{x} \underbrace{\ldots}_{y} \mid_{p} a_n$

↳ Finite Regular Languages ✗.

??

Context-Free-Grammars

G_1

$S \to 0S1$

$S \to R$

$R \to \varepsilon.$

Rules (Substitute)

Start variable is the top-left one.

Rule: Variable → Str. of variables and terminals.

↳ Left Hand Side. ↳ RHS

Strings Generation → 1. Write Start variable.

2. Replace any var. acc. to a rule.

Repeat this until only terminals remain.

3. $L(G_1)$ is the language of all generated strings.

S

↓

$0S1$

↓

$00S11.$

↓

00R11.

0011.

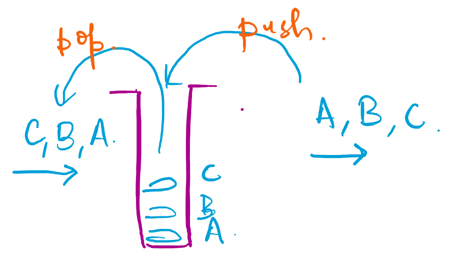
"Empowering ourselves"

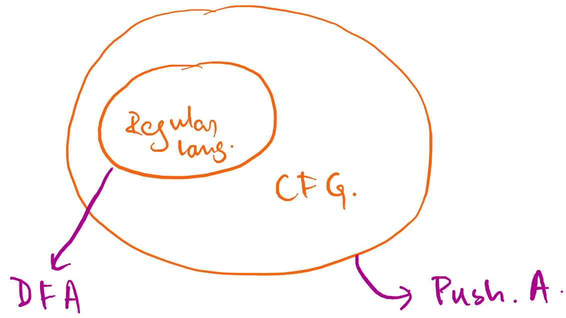
DFAs/NFAs. (finite memory).

PushDown Automata. (Stack).

Stack is a Data Structure.

FILO ↔ LIFO.



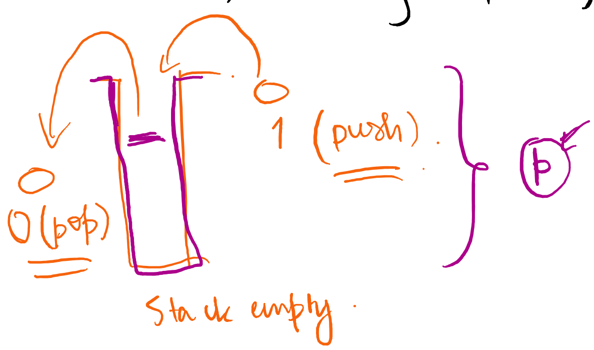


~~DFA C~~

Infinite. → Restrictions.
LIFO.

$L = \{ w \mid w \text{ has same no. of 1's and 0's} \}$

↳ Non-Regular Language.



PushDown A.

if #0 = #1
accept.
else
reject.

A CFG G is a 4-tuple (V, Σ, R, S) → Start var.

→ set of rules : $V \to (V \cup \Sigma^*)$ (finite)

→ terminal symbols

(finite)

Variables (finite)

terminal symbols (finite)

$S \to 0S11 0$

$\in (V \cup \Sigma^*)$

$V \to (V \cup \Sigma^*)$.

$L(G)$.

$u, v \in (V \cup \Sigma^*)$.

1. $u \rightsquigarrow v$ if we can go from u to v in one substitution step.

2. $u \overset{*}{\rightsquigarrow} v$. — — some no. of substitution steps.

$L(G) = \{ w \mid w \in \Sigma^* \text{ and } S \overset{*}{\rightsquigarrow} w \}$.

$(Q, \ldots, q_0)$ DFA.

$q_0 \in Q$.

$S \to 0S1$, $S \to R$ $\Big\}$ $S \to 0S1 \mid R$

Ex.

C_1: $B \to 0B1 \mid \varepsilon$.

$\times$ $B1 \to 1B$.

$0B \to 0B$

C_2: $S \to 0S \mid S1$

✓ $R \to RR$.

$\downarrow$

$L(C_2) = \emptyset$.

Next Class topics.

↳ Some other examples of CFGs.

↳ "Ambiguity" in CFGs.

↳ PushDown Automata.

↳ Thm. If A is a CFL then some PDA recognizes A.

$\longleftarrow$

↳

↳ Presentation. topics.